Tier 3 Finalized Testbed Architecture

& Folded Build Plan · Project CARS

High-fidelity Purdue-zoned ICS · Factory I/O + real S7-1200/1500 · full IT→OT attack path · July 2026

1. Decisions locked in

- **Fidelity: Tier 3** — full Purdue stack (L0–L5) with Industrial DMZ, corporate/IT zone, and dual firewalls, so attacks can traverse a realistic IT→OT kill chain.
- **Level 0 process: Factory I/O** (licensed on the university Dells) — 3D-visualized process driven by the real PLC via the Siemens S7 driver.
- **Hardware:** 2 Dell (full admin, 16GB+ each), 1 Asus (Win 11), real Siemens S7-1200/1500. MacBook M1 excluded.
- **Protocols:** S7comm (process I/O + realism), Modbus TCP (primary CARS traffic), OPC UA (L2–L3), MQTT/Sparkplug (IIoT edge).

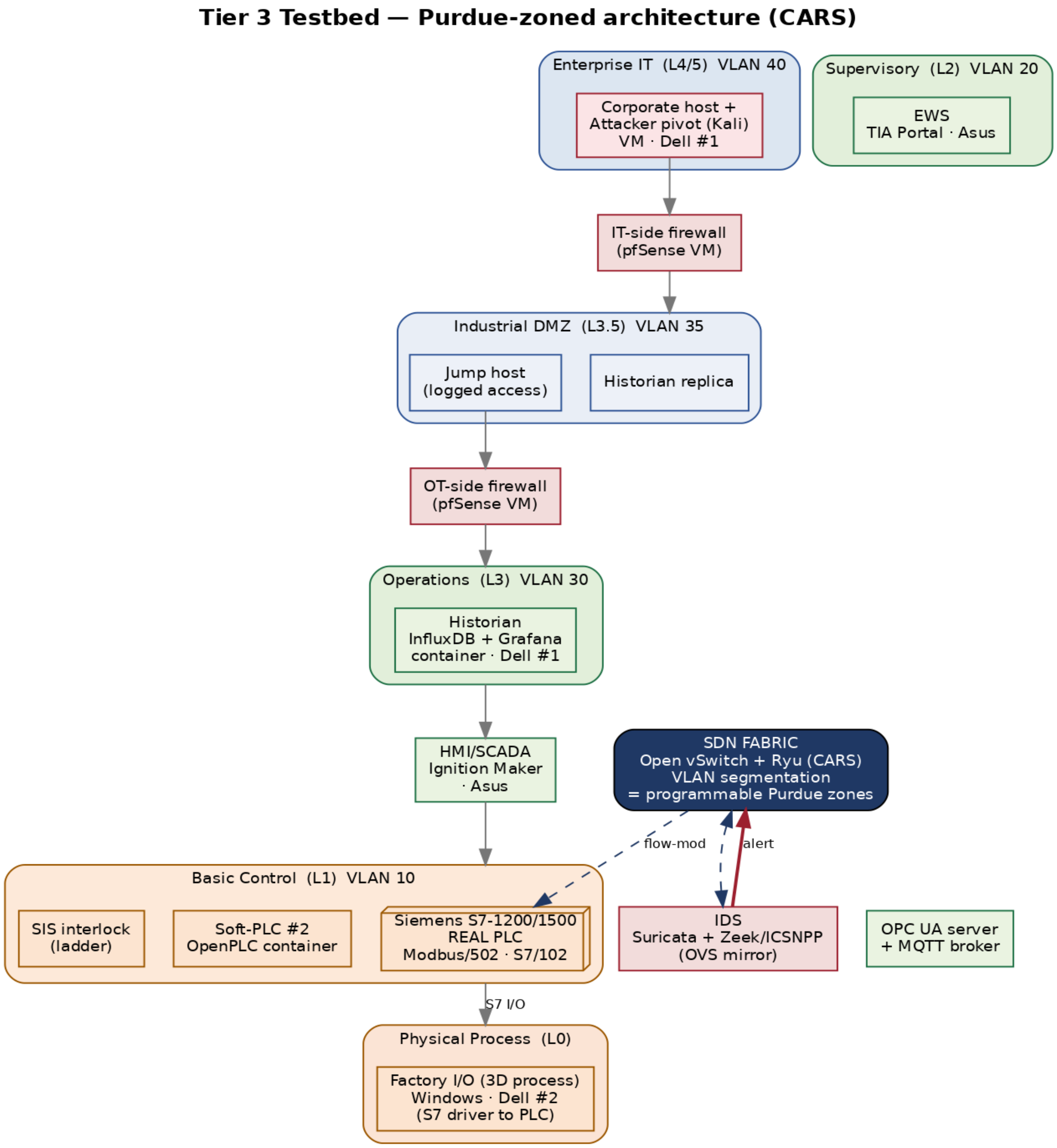
2. Host allocation

Factory I/O is Windows-only and GPU-bound, so it runs bare-metal on Dell #2 (Windows). Dell #1 boots Ubuntu and becomes the SDN core plus the virtualization host for every Linux role. The Asus is the operator/engineering station.

Host / OS	Runs	Purdue zone
Dell #1 · Ubuntu	Ryu + CARS + Open vSwitch (fabric); Docker/KVM host for IDS, historian, DMZ, firewalls, soft-PLC, OPC UA/MQTT, attacker	Network fabric + L1/L3/L3.5/L4-5 (virtualized)
Dell #2 · Windows	Factory I/O (3D process); optional offload of the Kali attacker VM	L0 Physical process (+ L4/5 option)
Asus · Win 11	HMI/SCADA (Ignition Maker Edition); TIA Portal (EWS)	L2 Supervisory
Siemens S7-1200/1500	Real control logic; Modbus server + S7comm; safe I/O; SIS interlock	L1 Basic Control + L0 real I/O
MacBook Air M1	Writing / notes only	— excluded

3. Zoned topology

Every device attaches to Open vSwitch on Dell #1. Purdue zones are realized as VLANs on OVS; two pfSense VMs enforce the IT ↔ DMZ ↔ OT boundaries with deny-by-default rules. The SDN controller (CARS) still mediates every flow and the IDS reads a mirror port — so the reactive loop operates across the whole zoned network.



4. Network / VLAN plan

VLAN	Zone	Members
10	Control (L1)	Real S7-1200/1500, OpenPLC soft-PLC, SIS interlock
20	Supervisory (L2)	HMI/SCADA (Ignition), EWS (TIA Portal), OPC UA server, MQTT broker
30	Operations (L3)	Historian (InfluxDB + Grafana)
35	Industrial DMZ (L3.5)	Historian replica, jump host — between the two pfSense firewalls
40	Enterprise IT (L4/5)	Corporate host + attacker pivot (Kali)
—	Process (L0)	Factory I/O ↔ PLC over the S7 driver (rides VLAN 10 to the PLC)
mgmt	SDN control	Ryu ↔ OVS OpenFlow channel; IDS mirror port (out-of-band)

Firewall posture: deny-by-default, allow-by-exception, fail-closed at each zone boundary — the standard OT segmentation model. CARS operates within and across these zones at the SDN layer, complementing (not replacing) the L3 firewalls.

5. Dell #1 RAM budget (fits in 16GB)

Component	Est. RAM	Note
Ubuntu host + Open vSwitch + Ryu/CARS	2.5 GB	Core; always on
IDS container (Suricata + Zeek/ICSNPP)	1.5 GB	Reads OVS mirror
Historian (InfluxDB + Grafana) containers	1.0 GB	Time-series tag logging
IT-side firewall (pfSense VM)	1.0 GB	Lightweight
OT-side firewall (pfSense VM)	1.0 GB	Lightweight
DMZ host (historian replica + jump)	1.0 GB	Container/VM
OpenPLC soft-PLC + OPC UA/MQTT containers	1.0 GB	Light
Kali attacker VM (during scenarios)	3.0 GB	Offload to Dell #2 to free RAM
Headroom	~4 GB	OS cache / spikes

Tip: prefer Docker containers over full VMs for the soft-devices (the ICSSIM / ICS-SimLab approach) — much lighter than VM-per-device. Run the Kali attacker only during Phase E, or host it on Dell #2 alongside Factory I/O, to keep the OT host comfortable.

6. Folded phased build plan (Tier 3)

The six phases from the build plan, updated with the Tier 3 components. Exit criteria and reading mappings from the master list still apply.

Phase A — Bench, process & baseline HMI

- Dell #1: install Ubuntu, Open vSwitch, Ryu, Docker; create br0 with VLANs 10/20/30/35/40; add physical ports via USB-Ethernet.
- Dell #2 (Windows): install Factory I/O; build a simple scene (e.g., level/tank or sorting station); configure the Siemens S7 driver to the real PLC.
- PLC: TIA Portal program that reads Factory I/O sensors and drives actuators + a safe LED/relay; enable PUT/GET and a Modbus TCP server; add a basic SIS interlock.
- Asus: install Ignition Maker Edition (HMI) and confirm it can read PLC tags via OPC UA/Modbus.
- **Exit:** Factory I/O process runs under real-PLC control, visible on the Ignition HMI, with all traffic through OVS.

Phase B — SDN baseline + historian + protocols

- Ryu learning-switch app manages OVS; verify control-loop latency/jitter within budget.
- Stand up the historian (InfluxDB + Grafana container) logging PLC tags; add OPC UA server + Mosquitto MQTT broker.
- **Exit:** Live process data flows PLC → historian and renders in Grafana; you can add/remove an OVS flow from Ryu.

Phase C — Zones, firewalls & detection

- Deploy the two pfSense firewalls and the DMZ host (historian replica + jump); enforce deny-by-default between VLANs 40 ↔ 35 ↔ 30.

- Deploy IDS (Suricata + Zeek/ICSNPP) on an OVS mirror; write Modbus/S7comm/OPC UA rules; emit alerts to the CARS Event API.
- **Exit:** An attacker in the enterprise zone is contained by the firewalls, and an OT-network anomaly reaches the Event API.

Phase D — CARS reactive engine (the contribution)

- Ryu CARS app installs block / mirror / redirect flows on demand; criticality model tags the PLC control loop vs unknown hosts.
- Safety guard: never hard-block the critical loop — mirror/redirect for inspection; block only untrusted/unseen devices; policy-check before install.
- **Exit:** Unseen device that alerts is auto-blocked; alert on the critical loop is redirected — Factory I/O process never interrupted.

Phase E — Full IT→OT attack scenarios & evaluation

- Kill chain from the enterprise zone: pivot → DMZ → OT; plus direct OT attacks (unauthorized Modbus/S7 writes, false-data injection, DoS).
- **Metrics:** detection-to-mitigation latency; process-safety preservation (watch the Factory I/O process); false positives; controller load. Log for reproducibility; benchmark vs SWaT/WADI/HAI conventions.
- **Exit:** A repeatable results table per scenario — the dissertation's core evaluation.

Phase F — Stretch

- P4 data-plane allowlist (bmv2); redirect-to-honeypot response; MiniCPS/digital-twin pre-validation of disruptive actions; MQTT/Sparkplug edge scenario.

7. Prerequisites & shopping list

- **Hardware:** 2–3 USB-to-Ethernet adapters for Dell #1 (to attach PLC, Asus, and uplinks to OVS). This is the only purchase.
- **Software (all free unless noted):** Open vSwitch, Ryu, Docker (Ubuntu); pfSense; Suricata + Zeek/ICSNPP; InfluxDB + Grafana; OpenPLC; Mosquitto; open62541/python-opcua; Ignition Maker Edition; Kali. Factory I/O (already licensed) and TIA Portal on Windows.
- **Verify on PLC:** PUT/GET enabled; "optimized block access" setting compatible with the Factory I/O S7 driver and Modbus server block configured.

8. Ready to build — first move

Start with the Phase A minimal loop before wiring the zones: get Factory I/O + real PLC + Ignition HMI working through OVS on a single flat network, then layer in the VLANs, firewalls, IDS, and CARS. Build the fidelity outward from a working core, exactly as the master reading list is ordered. Say the word and I'll write the concrete Phase A setup commands (OVS bridge + VLANs, Ryu install, Factory I/O S7 driver config, and the TIA Portal Modbus/PUT-GET settings).

This supersedes the tier selection in the component analysis; the phased build plan and master reading list remain the companion documents. Diagram zones are logical (VLANs on one OVS), not separate physical switches.